

STEP 1 — RETICULOCYTE INDEX

RI <2	Underproduction (marrow problem) → MCV + iron studies
RI ≥3	Overproduction → blood loss, hemolysis, sequestration

Hemodilution (aggressive IVF; hypervolemia/HF) drops Hgb without raising RI. RI may be falsely low if underproduction coexists with a bleed.

UNDERPRODUCTION — BY MCV

MCV	Causes
Macro >100	B12/folate, drugs (HU/allopurinol/HIV), liver/ ETOH, hypothyroid, MDS (esp. >120)
Normo 80–100	ACD, CKD (↓EPO), early Fe-def, acute blood loss, marrow (aplastic/leukemia/myeloma)
Micro <80	Iron deficiency, thalassemia, sideroblastic (Pb/ Zn/INH/linezolid), ACD

Other cell lines down → suspect primary marrow pathology.

IRON STUDIES

	Fe	TIBC	%sat	Ferritin
Fe-def	↓	↑	↓ (<20%; <10% strong)	↓ (<30 dx)
ACD	↓	↓/nl	↓/nl	nl/↑ (APR)
Thal	nl/↑	nl/↑	nl/↑	nl

Ferritin <30 = iron deficiency even in inflammation (inflammation only raises it). ACD + Fe-def coexist in 20–85% — clue is very low TSAT; use sTfR/hepcidin or an iron trial.
New Fe-def in older adult → **GI w/u** (colon CA until proven otherwise).

OVERPRODUCTION — HEMOLYSIS

Confirm hemolysis: ↑LDH, ↓haptoglobin, ↑indirect bili, + smear. Then the **DAT**:

DAT	Type / smear
Positive	Immune (AIHA): warm IgG (cancer/lymphoma, SLE, drugs); cold IgM (lymphoma, mycoplasma, mono; PNH, syphilis). Spherocytes
Negative	Non-immune: MAHA (schistocytes) vs non-MAHA

NON-IMMUNE (DAT-) BREAKDOWN

MAHA (schistocytes): + ↓plt → **TMA — emergency**. Primary TTP/HUS, drug-induced; secondary DIC (↓fibrinogen), HELLP; other — malignant HTN, prosthetic valve/LVAD/TIPS, scleroderma renal crisis.
Non-MAHA: intrinsic — membrane (HS, elliptocytosis, stomatocytosis, PNH), Hb (sickle, thal), metabolism (G6PD, PK); extrinsic — liver/hypersplenism, infection (malaria/Babesia/ Bartonella; Clostridium toxin), oxidant meds.

TMA / TTP — ACT FAST

Schistocytes + thrombocytopenia → **PLASMIC score**, send **ADAMTS13 BEFORE plasma exchange**, and start PEX if TTP likely (don't wait for the level). Confirmed TTP: **PEX + steroids + caplacizumab**. Avoid platelet transfusion in TTP.

Repletion: oral ferrous sulfate (alternate-day dosing improves absorption); IV iron (ferric carboxymaltose/ derisomaltose) if malabsorption, ongoing loss, CKD/HF, or oral intolerance.